AMRITA VISHWA VIDYAPEETHAM
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INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

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Team Name: Vortex

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Presentation Outline

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Problem Statement

Heart and lung screening is done separately, in clinics, with costly equipment. So most people in rural areas and in schools see no sign of smoking harm until symptoms appear, and no low-cost device ties smoking exposure, lung effort and heart strain into one score that a health worker can explain in a minute.

Problem Understanding

- **Who experiences this problem?** Rural communities, school and college students and smokers with no easy access to clinic screening. About 270 million Indian adults use tobacco, about 200 million of them in rural areas, and about 100 million smoke (GATS-3, 2025–26).
- **What causes the problem?** Heart and lung checks use separate, costly devices in clinics. A standard pulse oximeter cannot tell CO-bound from oxygen-bound haemoglobin, so SpO_2 can look normal even when CO is high.
- **How does it affect users or stakeholders?** Smoking and secondhand smoke kill about 1 million Indians a year. Cardiovascular disease caused 28% of deaths in 2016, and India has 32% of global chronic respiratory DALYs. The COVID-19 second wave (April 2021) showed that people need a cheap screening tool before an emergency.

Existing Gaps & Challenges

Feature	Smokerlyzer	Clinical spirometer	Pulse oximeter	ECG machine	Heart-Lung Check
CO smoke marker	Yes	No	No	No	Yes
Breath flow	No	Yes	No	No	Screening-grade (peak flow + 1-s volume index)
SpO_2 and heart rate	No	No	Yes	Heart rate: Yes SpO_2: No	Yes
Single-lead ECG	No	No	No	Yes (if supported)	Yes (rhythm flag only)
One explainable score	No	No	No	No	Yes (Green/Yellow/Red + driver)
Biomass-context flag	No	No	No	No	Yes
Offline rural use	Partial	Yes	Yes	Depends on model	Yes
Indicative price	₹40k–70k	₹50k–1.5L	₹1k–3k	₹15k–40k (varies by model)	₹14,500 (BOM ₹9,000)

- **Main challenge that remains unsolved:** no low-cost device ties smoking exposure, lung effort and heart strain into one score a health worker can explain in a minute.
- **Added challenge:** exhaled CO can also come from cooking smoke or traffic exhaust. Standard smoking tools miss solid biomass fuel smoke, which affects over 40% of Indian households, so a rural cook should not automatically be labelled a smoker.

Innovation & Uniqueness

New

to our knowledge, the first low-cost device combining a smoking marker (CO), fingertip heart and oxygen readings, and a breath-effort index in one offline score.

Why it suits the problem

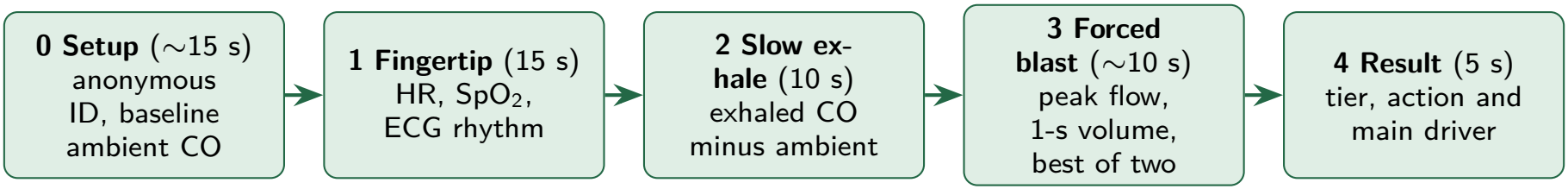
the CO channel adds what an oximeter is blind to, and all channels are read in the same 90 seconds.

Differentiator

an explainable score that shows why it is yellow or red.

Our Proposed Solution

- **Core idea:** Heart-Lung Check, one portable ESP32 device and one test of under 2 minutes, giving one clear result that tells a health worker what to do next.
- **How it works:** a fingertip sensor (heart rate, SpO_2, ECG rhythm), an exhaled-CO chamber and a flow sensor combine into a Green / Yellow / Red result with the main driver, fully offline.
- **Main value delivered:** an explainable action (counsel, recheck or refer), not a diagnosis, at a fraction of the cost of three separate devices.



Data collected from hospitals and clinics is compared with the regression model's predictions, and the model is fine-tuned to improve its accuracy.

GREEN · Low risk
Healthy habits advice, routine follow-up

YELLOW · Moderate risk
Counselling and a recheck in a few weeks

RED · High risk
Refer to PHC for spirometry and a doctor

Biomass-aware: CO from cooking smoke is flagged as exposure context, so a rural cook is never labelled a smoker by default.

Technical Approach

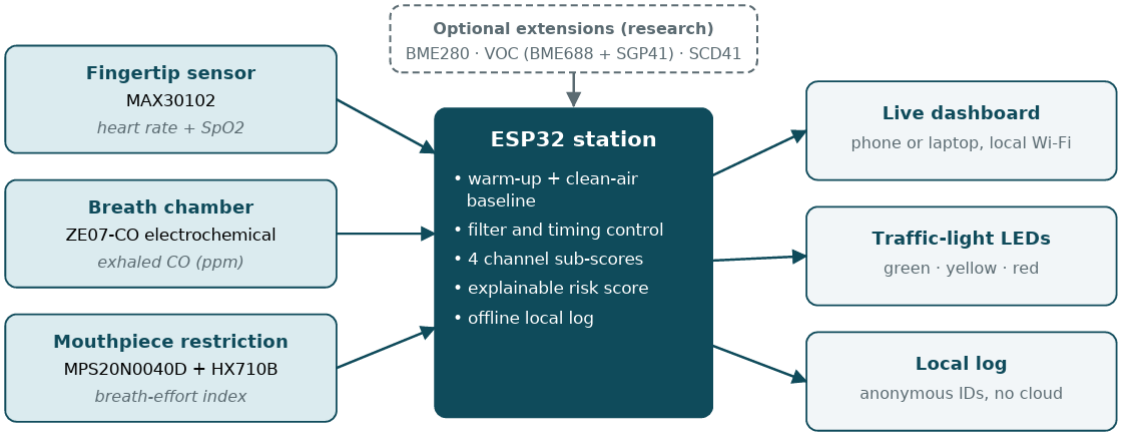


Figure 1. System block diagram. Everything runs locally; no internet is needed.

Three sensors feed the ESP32, which builds four channel sub-scores into one explainable risk score shown on a dashboard and LEDs.

Tech Stack

Frontend: live dashboard (phone or laptop, local Wi-Fi); traffic-light LEDs	**Database:** on-device local log, anonymous IDs, no cloud
Backend: ESP32 firmware: warm-up, clean-air baseline, filtering and timing, sub-scores	**AI/ML, APIs, Hardware:** explainable risk score from four sub-scores; MAX30102, ZE07-CO, MPS20N0040D + HX710B (pro tier: MAX86150, ME3-CO + LMP91000, SDP810)

Feasibility & Scalability

- **Feasibility:** off-the-shelf sensors and an ESP32, with a prototype cost of about Rs. 3,500. A 3D-printable Venturi tube and a simple, explainable model keep it buildable locally. Phase 1 proves CO repeatability on the bench against a smokerlyzer, then a pilot with 30+ adult volunteers validates against reference devices.
- **Scalability:** one campus, then pilot villages, one PHC and schools and camps around Coimbatore, then NGO and workplace camps and PHC partners, with a district rollout using a consent-based dashboard, then other states with regional-language UIs. Same hardware, thresholds tuned per region; each unit works offline, at up to about 40 people an hour.
- **Risks:** sensor drift and cross-sensitivity are handled with a warm-up, a clean-air baseline (ambient subtracted), recalibration, a known-CO check and confounder questions. Weak blows are handled by a quality gate and best of two. Green means "no flags today", not "healthy", and a health worker stays in the loop. Claims stay at awareness screening until a Medical Devices Rules 2017 review.
- **Validation roadmap:** (1) bench build, each channel checked alone; (2) calibration of CO against a reference and flow against a calibrated syringe or spirometer; (3) pilot data from smokers, non-smokers and biomass households, with spirometry as the reference to train the model; (4) ASHA/ANM and teacher usability field trial, then refine thresholds. Targets: under 2 minutes per screening, $\geq$90% valid tests within two attempts, sensitivity and specificity reported against spirometry.

Cost for Product

What the BOM covers (about Rs. 9,000)

- ESP32 and PCB
- MAX86150, CO sensor, SDP810, BME280
- Display, Li-ion battery, enclosure
- 3D-printed Venturi tube
- Consumables per test: mouthpiece and filter

Indicative price (midpoint of range)

Smokerlyzer + spirometer + oximeter

Rs. 1,57,500(approx.)

Heart-Lung Check (target price)

Rs. 14,500

- **Development or hardware cost:** target price Rs. 14,500 per unit with a bill of materials of about Rs. 9,000. The bench prototype tier costs about Rs. 3,500 (ZE07-CO is Rs. 970 on a retail listing; other parts are estimates to be confirmed with quotes).
- **Hosting or maintenance cost:** no cloud hosting, since everything runs locally. Running costs are the disposable mouthpiece (Rs. 2–5) and filter, and sensor upkeep of about Rs. 0.20 per screening.
- **Estimated total:** Rs. 14,500 per unit, about 6–15× cheaper than buying the three separate devices (smokerlyzer Rs. 40k–70k, spirometer Rs. 50k–1.5L, oximeter Rs. 1k–3k), while adding ECG, the biomass flag and one score.

Use Cases

Use case 1: school or college health camp

A teacher or health worker screens students (guardian consent for under-18s) and explains the colour result.

Use case 2: rural community camp or ASHA/ANM visit

Adults are screened, and red results are referred to the nearest PHC.

Use case 3: tobacco-cessation programme

Counsellors use the CO reading as an objective number and re-check after a few weeks.

Use Cases (continued)

Use case 4: PHC triage

A first-line check to decide who needs spirometry or a doctor.

Use case 5: NGO and workplace camps

Fast screening for field teams, and for factory and construction workers.

Use case 6: biomass households

Spot early risk in women and children who cook on solid fuel, with CO from cooking smoke flagged as exposure context rather than smoking.

Impacts and Benefits

Health

- Earlier detection before symptoms
- Fewer missed cases
- Smarter referrals

Social

- Reaches rural women and children
- Builds awareness of early heart and lung risk

Economic

- 6–15× cheaper than separate devices
- Fewer unnecessary clinic trips

System

- Data for district planning
- Consent-based, anonymous IDs

One screening, one clear next step: counsel, recheck or refer, in under 2 minutes.

Measurable targets (to prove in testing): under 2 minutes per screening; $\geq$90% valid tests within two attempts; sensitivity and specificity reported against spirometry; SpO_2 within $\pm$2%, heart rate within $\pm$3 bpm and CO within $\pm$2 ppm of reference devices; up to about 40 people per hour per station.

Sustainable Development Goals (SDGs)

þSDG Number	þTarget Number	þGoal Name	þConnection
SDG 3	Target 3.4 (Non-Communicable Diseases)	Good Health and Well-Being	Detects early-stage cardiovascular disease and chronic respiratory conditions (COPD, lung impairment) before severe symptoms develop, addressing premature deaths from non-communicable diseases.
SDG 3	Target 3.9 (Hazardous Chemicals & Pollution)	Good Health and Well-Being	Addresses household air pollution from solid biomass cooking fuel, which affects over 40% of Indian households (mainly rural women and children).
SDG 3	Target 3.c (Health Financing & Workforce)	Good Health and Well-Being	Equips frontline ASHA and ANM workers with a portable, low-cost screening tool for doorstep visits.
SDG 10	Targets 10.2 & 10.3 (Equal Access to Healthcare)	Reduced Inequalities	Bridges the gap between urban specialised clinics and underserved rural communities by replacing costly, clinic-dependent equipment (Rs. 50k–1.5L) with an affordable screening station (Rs. 14,500).

Individual Member Contribution

þMember	þContribution or responsibility
Aruhi Bourasi CB.EN.U4ECE26307	Hardware and sensor integration
S. Vinay Siddharth CB.EN.U4ECE26341	Firmware and signal processing
Gayatri Girish CB.EN.U4ECE26314	ML model and scoring logic
Sriguru A CB.EN.U4ECE26350	Research, design and presentation

Team Vortex: four ECE students working together on every stage, from sensors to the final score.

Yukti ID

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